→ define the size of the cut pieces..

→ coordinates contained in patches.

→ then think about the data alot...

@ Data first

@ Handy diagrams to work with system.

@ get some go around the papers for the solutions.

Project Methodology Analysis

Q) Can Classification work for the given current use-case?

- Classification gives the 256×256 patch containing bacteria, but not where within that patch.
- Classification ranks patches, not individual bacteria.
  ↳ If patch has 5 bacteria?
- Patches are too large

Will a hybrid appraach work?

- 256×256 patches. Classification → get the positive patches.
- Divide into smaller 32×32 sub-patches. → Classify
- Draw bounding box around the sub-patches.
  → 32×32 still very Large. + Very slow?
- Will work if needed to highlight 10 most probable region.

- Classification identifies regions that might contain bacteria, but we need to show the exact loc of the bacteria. Object Detection provides bounding boxes and allow us to rank by confidence.

256×256 Classification

- Large enough to see gastric gland
- Small enough to maintain bacteria detail.

Prob: Pylori is about 5-15 pixels at 40x

- 256×256 is still relatively huge.

1024×1024 for NuClick segmentation.

- Needs more context for identifying complete gastric gland boundaries.

- 640×640 for YOLO
- Default i/p size for YOLO.
- Will divide into grids. (80×80, 40×40, 20×20)
- Capture more gastric gland structure

↓ patch decreases information

↑ increases cost

Solve Duplicate issue!! → Non-Maximum Suppressions (NMS)

→ Sort by confidence → take highest confidence

$IoU(A, B) = 0.72$ → Duplicate ($IoU > 0.5$) Remove

$IoU(A, C) = 0.05$ → Diff ($IoU < 0.5$) Keep

Repeat.

Retrospective of old App vs New Approach

* Pre-processing

HSV + OTSU

Converts WSI → HSV color space and uses Otsu's thresholding on the Saturation channel to create a mask.

→ Removes 60-70% useless data.

* Sliding window

→ Considers the entire image, instead of segmenting the gastric gland area. Recreating 1:100 imbalance prob.

* Dual Mode.

Highly redundant and confusing.

* Zoom-in classification.

If a 320x320 patch is positive, it suggests cutting it into 32x32 sub-patches and running again →

* Co-ordinate mapping.

Recording global co-ordinate, and adding local (x, y) from model detection.

Proposed new Plan

* U-Net / NuClick on low resolution thumbnail to find regions that matter: gastric glands.

Patch (512x512)

Detection (YOLO)

Mapping: Map co-ordinates back to WSI for final o/p.

Problem Statement? => CIFAR-100

* 22 WSI slides (veterinary gastric biopsies, 40x magnification)
* H&E stained. → Annotations from expert.
* Class imbalance and incomplete annotations.

Claude recommended Approach.

WSI → Sliding window (640x640) → YOLO → NMS → Ranking

Step 1: Sliding window Approach. → Move magnifying glass across an image.
* Large WSI cannot fit into GPU.
* Dividing into smaller patches → 640x640 → i/p for YOLO

Stride → no of pixels to move window each step.

If 20% overlap, then stride = 512. [128 overlap]

What if, there is no overlap?

If bacteria present at the edge of the patch? Half in patch 1 and other half in patch 2. => YOLO might not detect either.

Calculating no of patches in sliding window.

Start: $x = 0$

End: $x + 640 \leq 187,248$

No of patches = (Last pos - First pos) / Stride + 1

≈ 365 Horizontal

112 Verti

(Backward detection algorithm

* Convert RGB to grayscale.

↓

Count tissue pixels

tissues = pixels with val < 240 (darker than white)

bg = pixels with val ≥ 240 (white)

↓

Calculate tissue percentage

% = [(tissue_pixels) / (640x640)] * 100

If tissue % > 10%. → keep patch

else skip patch.

↓

Extract patches with co-ordinates.

↓

YOLO

↓ o/p

Convert to global co-ordinates

↓

But since there is 20% overlap, there is chance of duplicate detections.

Solve Duplicate issue!! → Non-Maximum Suppressions (NMS)

→ Sort by confidence → take highest confidence

IoU(A, B) = 0.72 → Duplicate (IoU > 0.5) Remove

IoU (A, C) = 0.05 → Diff (IoU < 0.5) Keep

Repeat.

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YOLO Pre-trained

* Medico YOLO → Pre-trained on Polyp detection in endoscopy.
* BreastNet YOLO → Breast cancer detection.
* Retinal YOLO → Specialized for diabetic retinopathy detection.
* Cell-YOLO: Trained on various cell detection tasks.
* ImageNet YOLO → what we are using.

YOLO → Where are bacteria present in this image?
↓
Get the co-ordinates + confidence.

What happens without negative samples?

* YOLO puts bounding boxes over everything which looks like a bacteria.
  eg. purple cells, curved structures in tissue.

Negative patches → These other structures are normal, do not detect them as positive.

* YOLO Background class is required.

If this happens, model thinks there should be a bacteria in every image.

Now, for active learning,
Training set ~50:50 → Mostly correct +ve/-ve distinctions.
less confusion.

↓
Train the model using this
↓
Use the model on the remaining
↓
Expert Review.
↓
Update dataset. → Retrain with increased patches.

PathAI:

* Mark a few example regions. Train classifier on
↑
segmentation.

YOLO Working

* Each patch (640 x 640) is divided into 80 x 80 pixels

↓

In each grid, it is responsible for detecting objects whose center falls in that cell.

↓

Each cell makes predictions

Usually 3, 4 → But very low confidence.

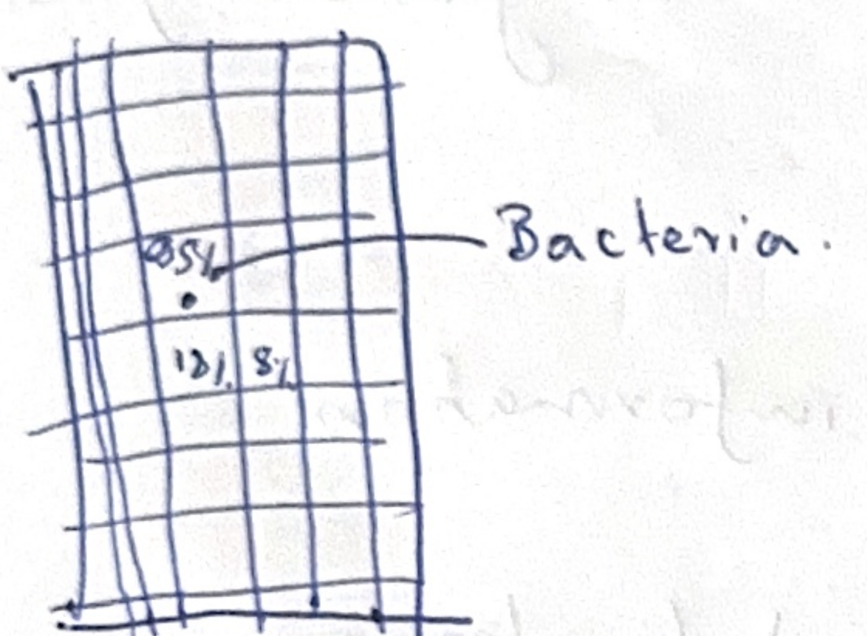
↓

Filter Low Confidence Predictions [threshold]

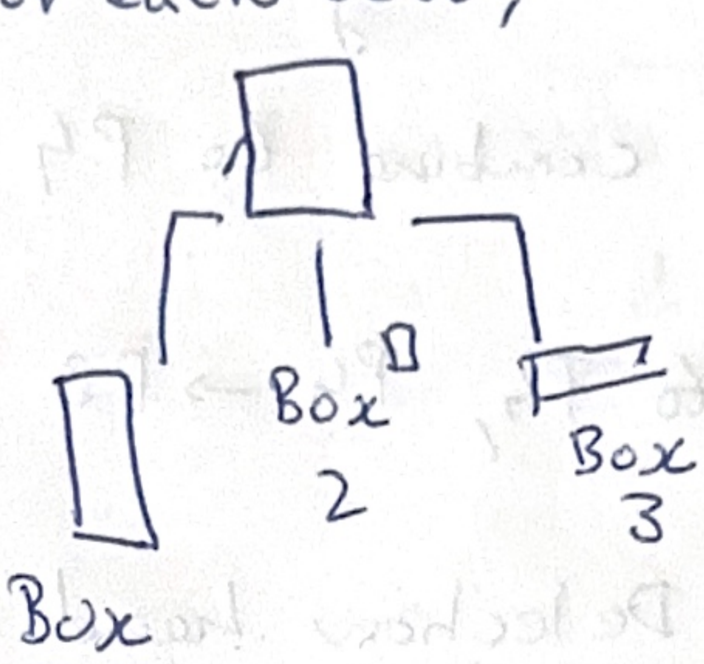
↓

NMS [If multiple boxes detect same bacteria]

First, YOLO divides image into grid, let this be 8 x 8. So 64 cells total



→ For each cell,



Box 1 → 95% confidence
Box 2 → 19%
Box 3 → 12%

All cells make prediction, but only one cell gets.

□ → Layer 1 (See edges) → Layer 2 (See shapes) → Layer 3 (See bacteria) → Detection (Found it!)

* During training, model splits into patches, YOLO then tries to see If bacteria present, mostly all cells give 2, 3 detections, but with lower confidence. If there is detection with higher confidence, it is taken.

After detecting, compares with ground truth, calculates location loss, Class loss and confidence loss and then adjusts weights.

Architecture.

I/p Image → 640x640x3

↓

Backbone

P1 320x320x64 } Edges, lines

↓

P2 160x160x128 } Corners, shape

↓

P3 80x80x256 } Spiral patterns

↓

P4 40x40x512 } Bacteria

↓

P5 20x20x1024 } Global context.

features

CSP Block

I/p features

↓ Part 1 → Conv 1 (Learns features)

↓ Part → Identity (skip connection)

↓

Concat

↓

Final Conv

↓

O/p

↑↓ computation

Better gradient flow

Neck → 20x20x1024 too coarse for detecting tiny bacteria. Need to bring back spatial detail.

Feature Pyramid Network

Upsample and combine to P4

P5 info passed to P4, P4 → P3. Sharing information.

80x80x256 → Detection head → Detect bacteria.

Active Learning Approach.

We have 3 well-annotated WSI Images.

2 → training, 1 → Validation.

- Keep 50-50 split → Positive patches : [Negative far from annotations, clear non-bacteria regions]

↓

Baseline Model.

↓

Use model to annotate 2/3 other WSI's.

↓

Predictions > 0.75 → High quality

↓

Send for verification.

- Negative patches
  - Patches > 500 pixels away from annotation
  - Background/white space regions.

CSP Block

I/p features

Part 1

Part 2

Identity (skip connection)

Concat

Final Conv

O/p

computation

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via region)

Cross-Validation.

- Dataset is fixed? Dataset grows over time. It will be hard to verify the results with Philipp.

- Do not add verified patches to Validation Set!

Label Studio → Keep track of Active Learning pipeline.